

Valentine Magic

It's Valentine's Day in Russia today, as we all know number of girls in Russia is more than number of boys hence boys have an extra advantage while choosing girl for the valentine evening. Each boy has certain number of chocolates and each girl has certain number of candies. Now you being the anchor of evening wants to pair all the boys with girls such that the sum of absolute difference between boy's chocolate and girl's candy in a pair is minimized. Ofcourse some of the girls will remain unpaired but that's okay as we are in Russia

Input format

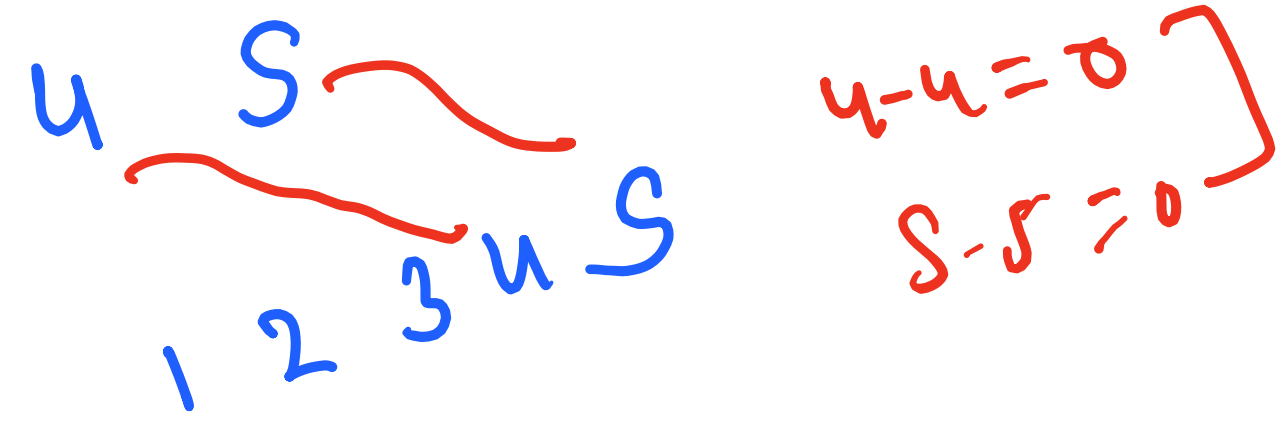
The first line consists of two integers N and M.
then follow N integers in second line.
Then follow M integers in third line.
M >= N

Output format

The only line which consists of required sum of absolute differences.

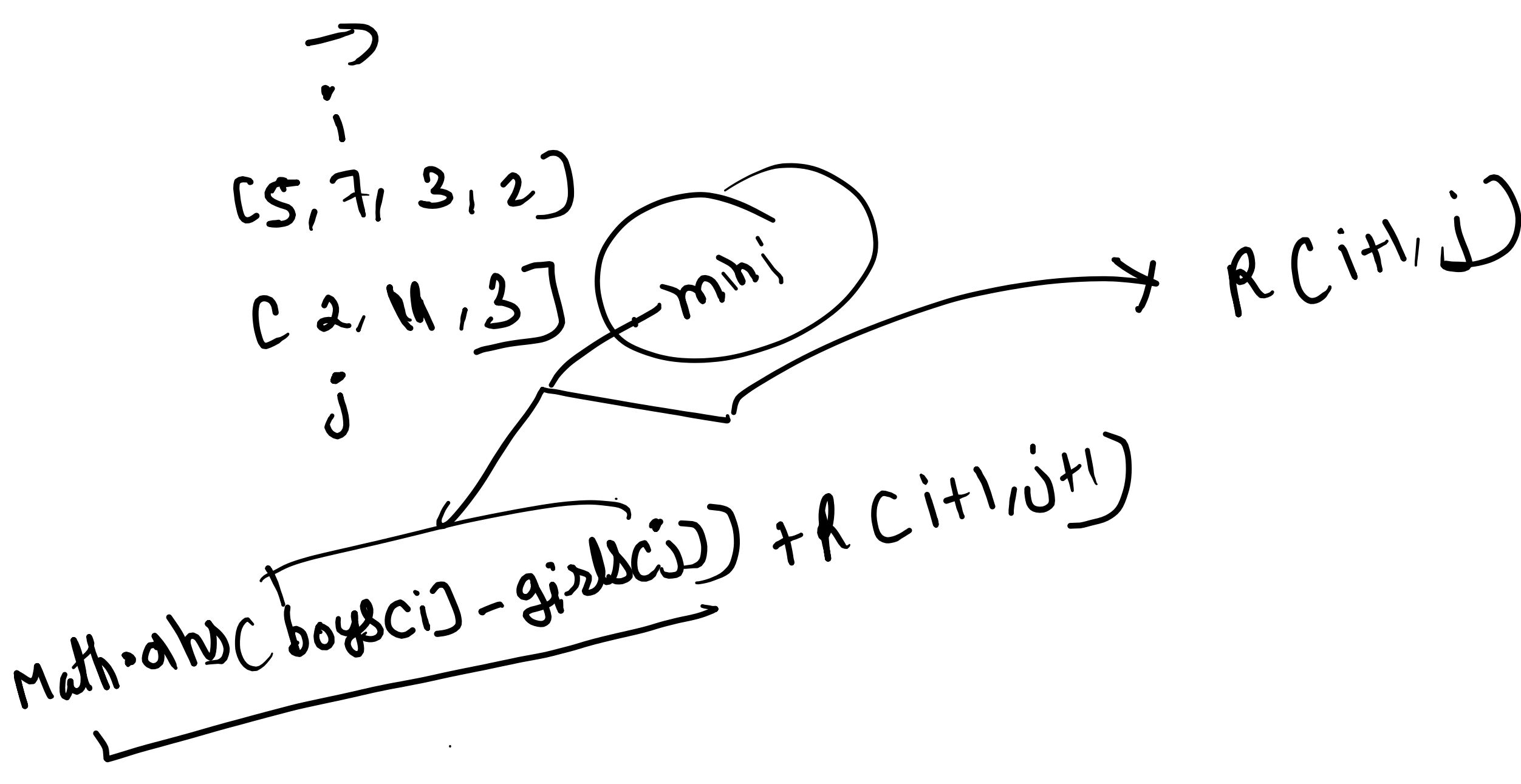
Example 1

Input
2 5
4 5
1 2 3 4 5



[5, 7, 3, 2] → girl
[2, 11, 3] → boy

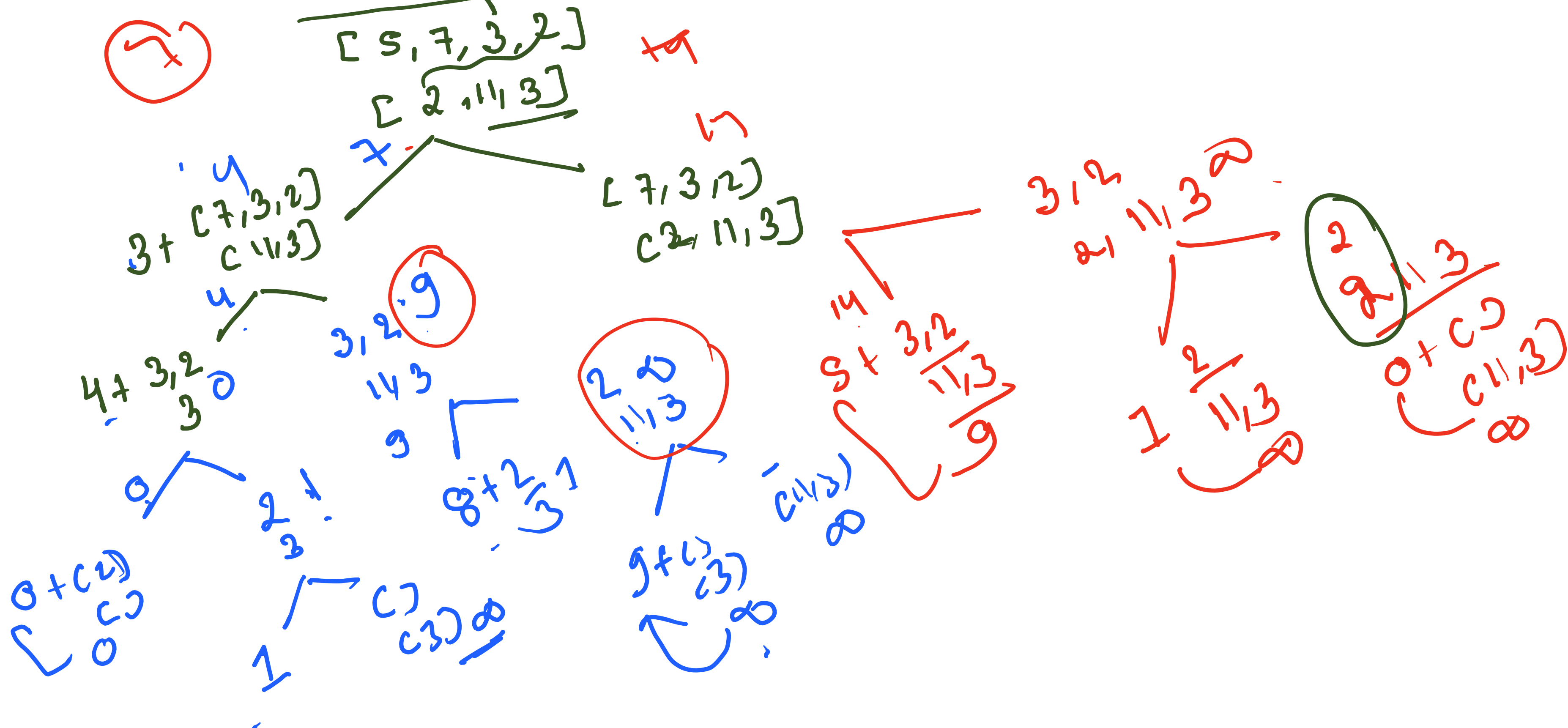
2-2=0
3-3=0
11-7=4



```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int[] boys = { 2, 11, 3 };
    int[] girls = { 5, 7, 3, 2 };
    System.out.println(Mini_Diff(boys, girls, 0, 0));
}

public static int Mini_Diff(int[] boys, int[] girls, int i, int j) {
    if (i == boys.length) {
        return 0;
    }
    if (j == girls.length) {
        return 8989079;
    }

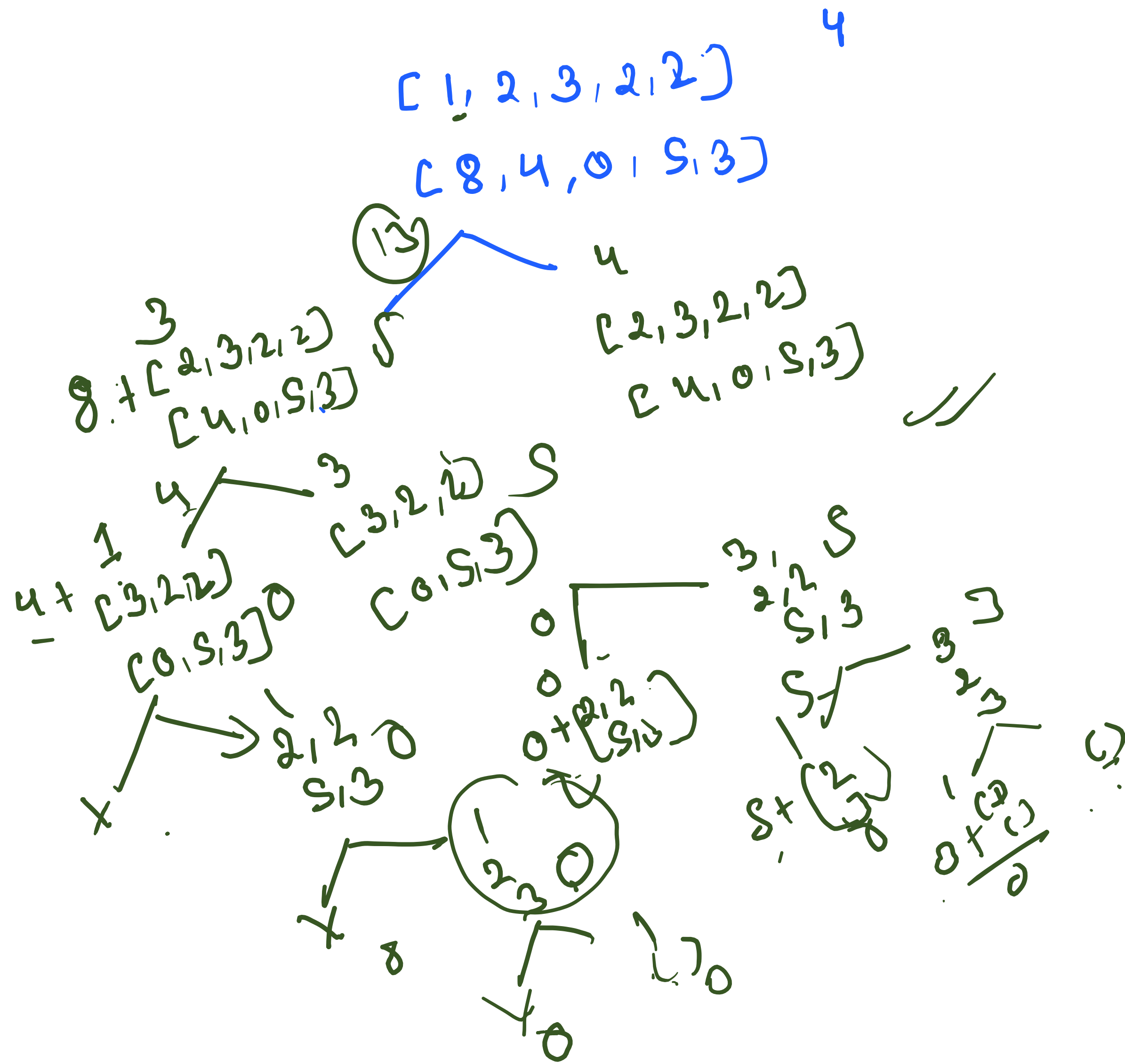
    int pair = Math.abs(boys[i] - girls[j]) + Mini_Diff(boys, girls, i + 1, j + 1);
    int nopair = Mini_Diff(boys, girls, i, j + 1);
    return Math.min(pair, nopair);
}
```



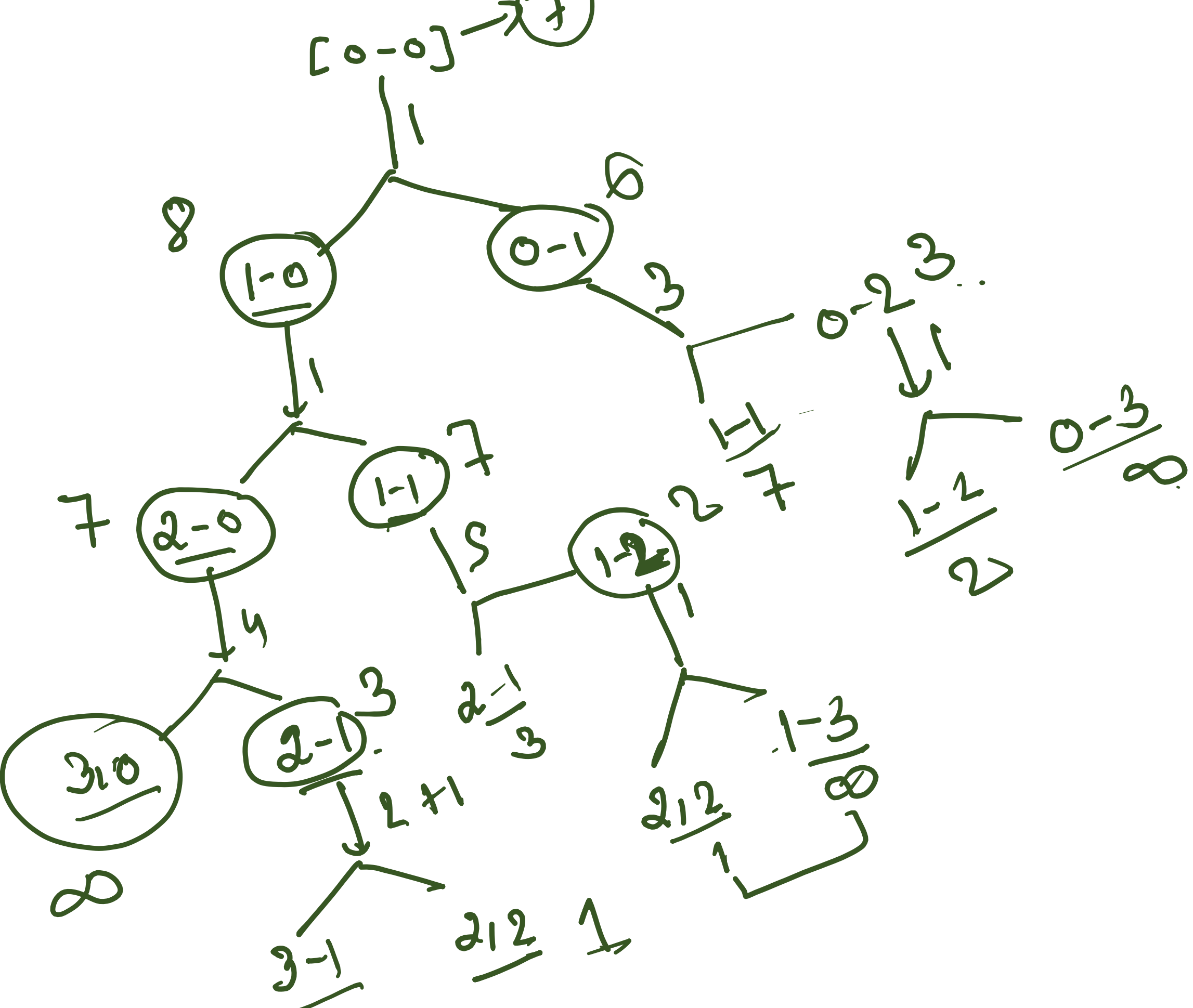
You are packing for a vacation on the sea side and you are going to carry only one bag with capacity S (1 <= S <= 1000). You also have N (1<= N <= 1000) items that you might want to take with you to the sea side. Unfortunately you can not fit all of them in the knapsack so you will have to choose. For each item you are given its size and its value. You want to maximize the total value of all the items you are going to bring. What is this maximum total value?

8 ✓
9
12 ✓
13 ✓
15 ✓

wt = [1, 2, 3, 2, 2]
val = [8, 4, 0, 5, 3]

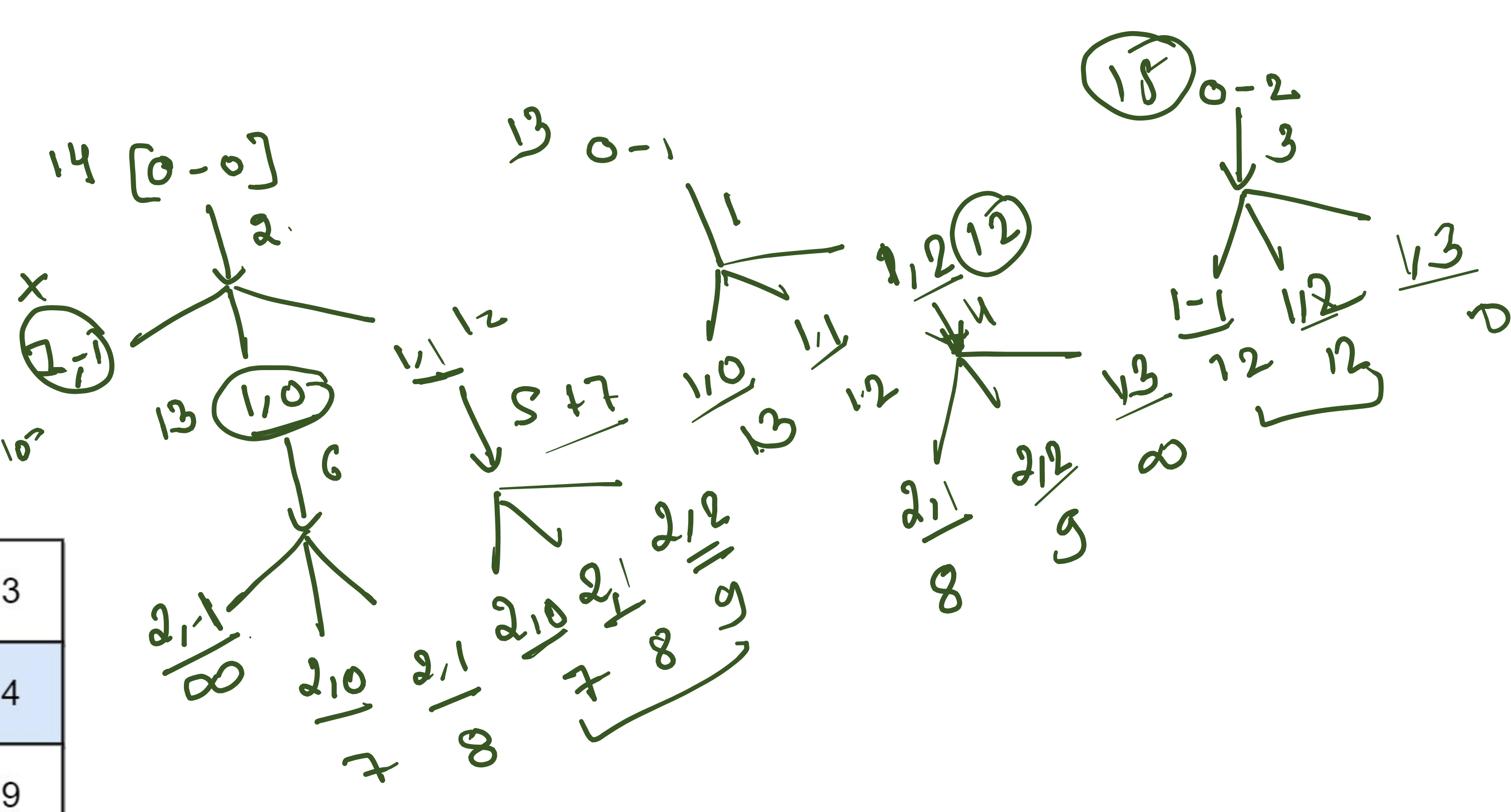


	0	1	2
0	1	3	1
1	1	5	1
2	4	2	1



2	1	3
6	5	4
7	8	9

2	1	3
6	5	4
7	8	9



1	2	3
4	5	6
7	8	9

